SYDE 556/750

Simulating Neurobiological Systems
Lecture 7: Temporal Basis Functions

Chris Eliasmith

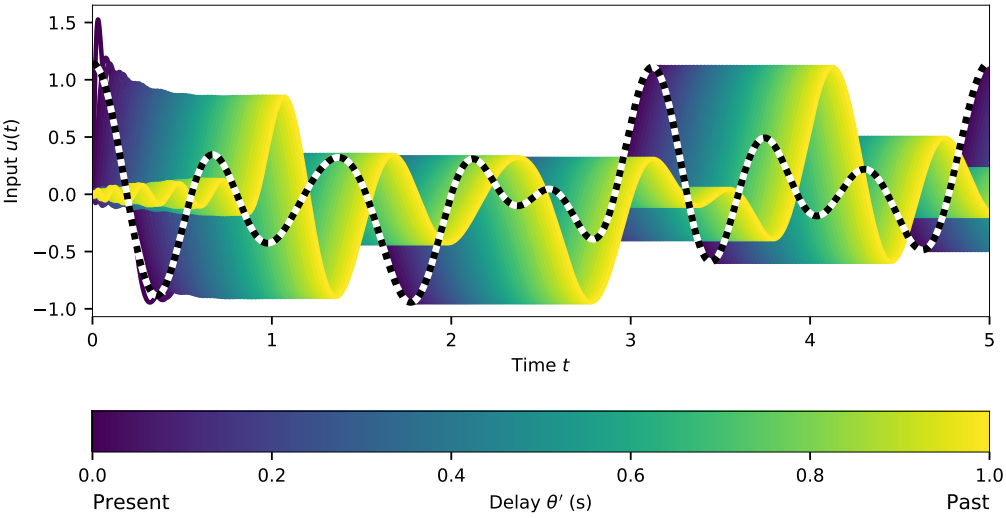
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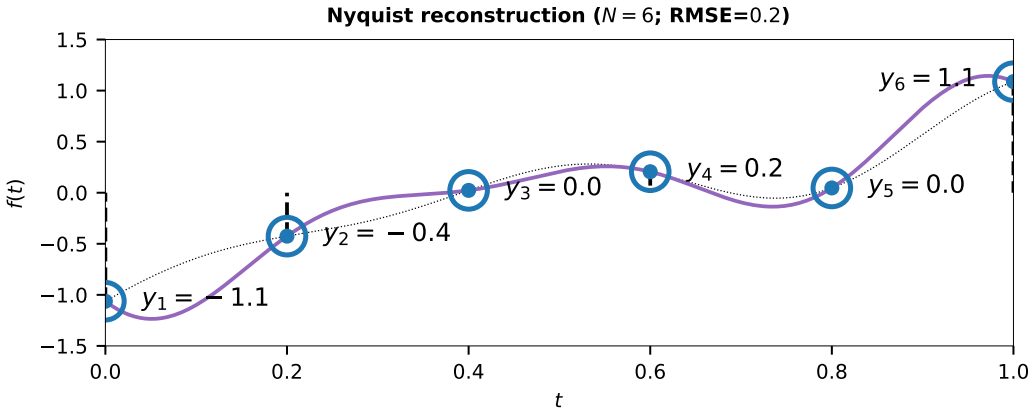


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Representing Stimulus Histories



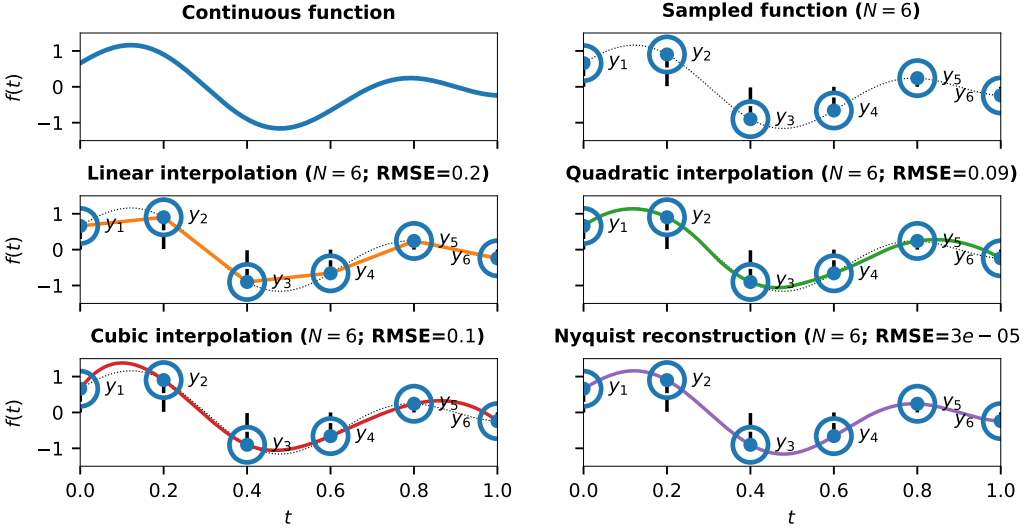
Representing Functions: Sampling



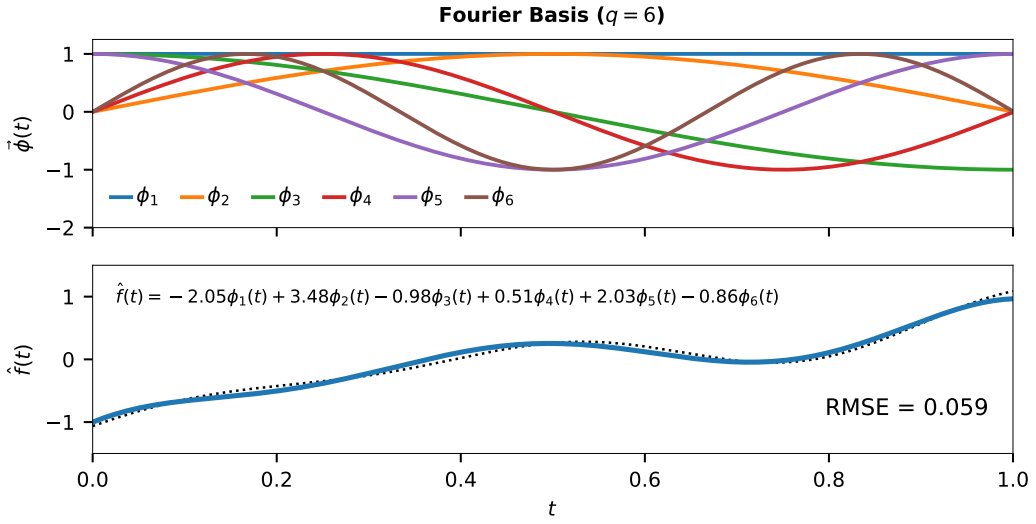
Nyquist Sampling

The Nyquist-Shannon Sampling Theorem
If $f(t)$ contains no frequencies greater than B then it is *completely* determined by samples spaced $\Delta t = \frac{1}{2B}$ apart ($N = 2BT$ equally spaced samples for a time-slice $[0, T)$). There is a *one-to-one mapping* between the samples $\mathbf{x}$ and the function $f(t)$.

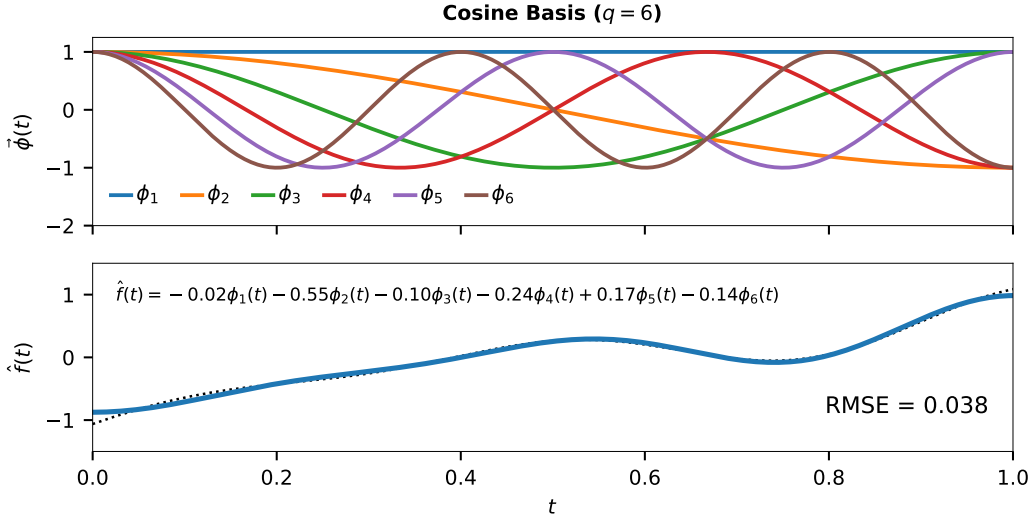
Representing Functions: Sampling <3Hz



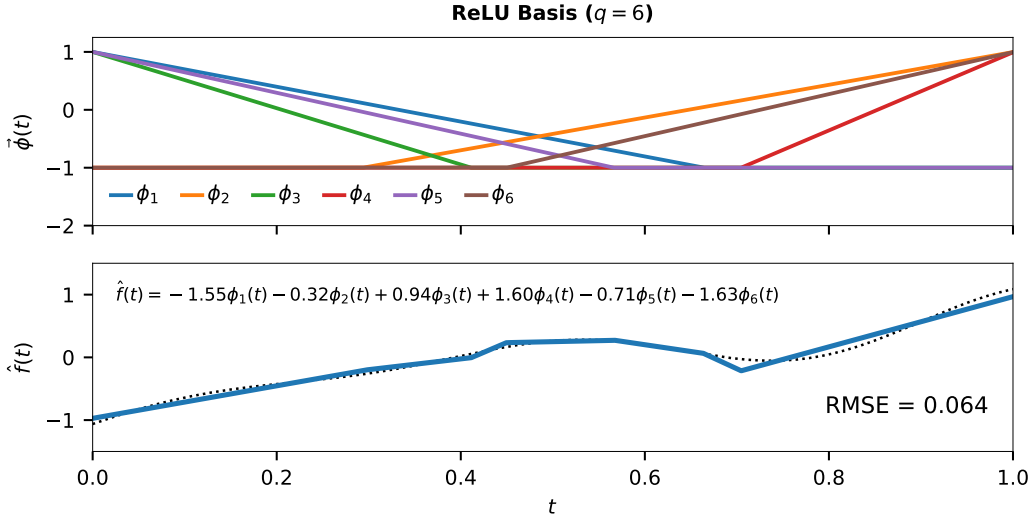
Representing Functions: Fourier Basis



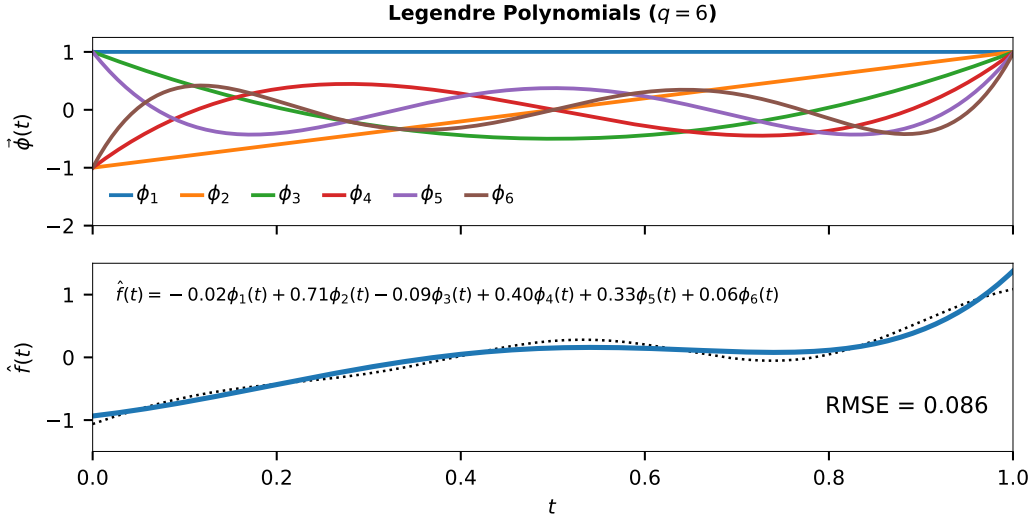
Representing Functions: Cosine Basis



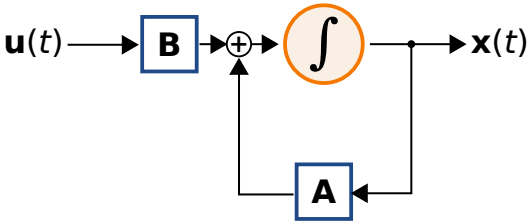
Representing Functions: ReLU Basis

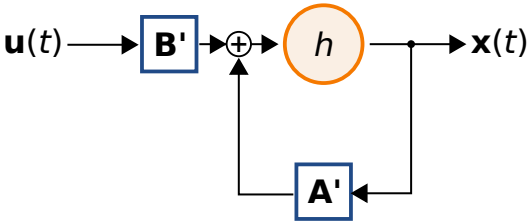


Representing Functions: Legendre Basis



Implementing the Delay Network



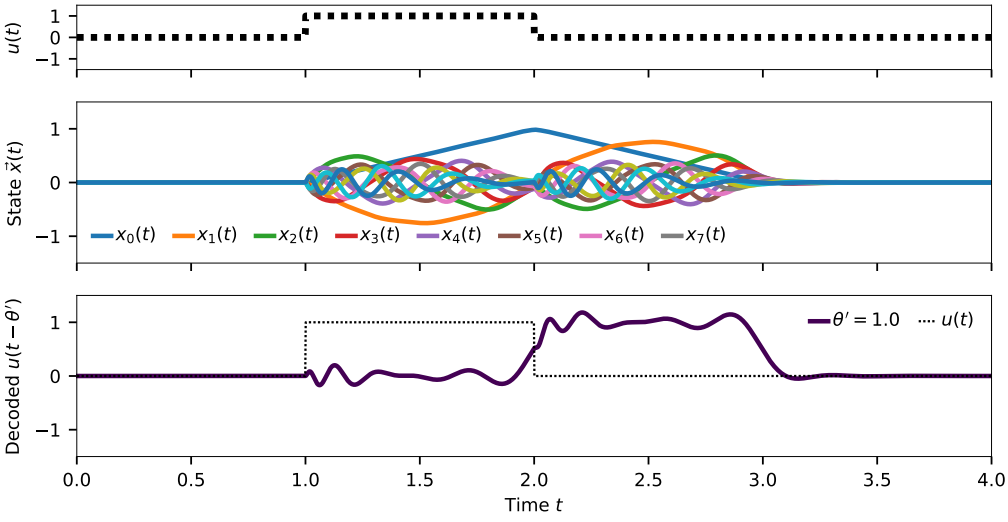


$$\theta\mathbf{A} = a_{ij} \in \mathbb{R}^{q \times q}, \qquad a_{ij} = \begin{cases} (2i+1)(-1) & i < j, \\ (2i+1)(-1)^{i-j+1} & i \geq j \end{cases}$$

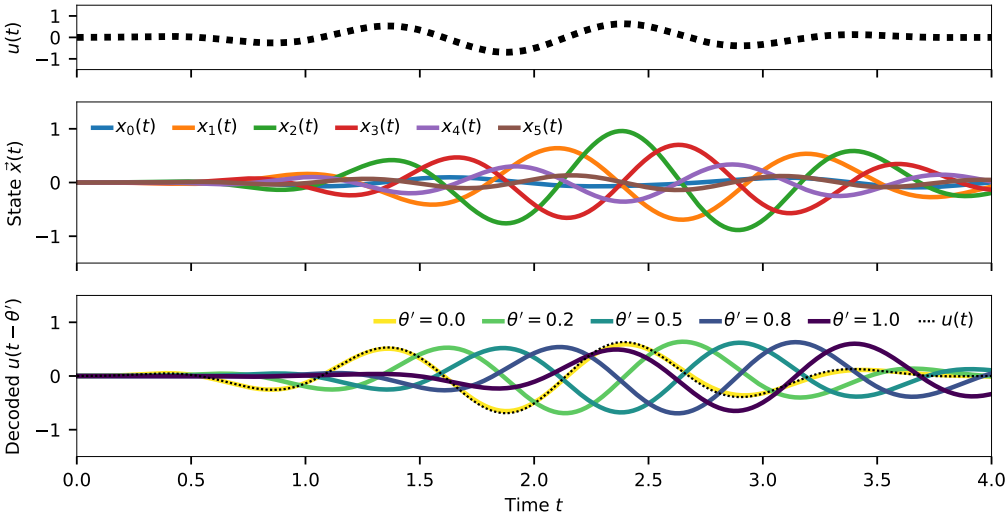
$$\theta\mathbf{B} = b_i \in \mathbb{R}^{q}, \qquad b_i = (2i+1)(-1)^i$$

$$\mathbf{A}' = \tau\mathbf{A} + \mathbf{I} \qquad \mathbf{B}' = \tau\mathbf{B}$$

Delay Network: Step Function

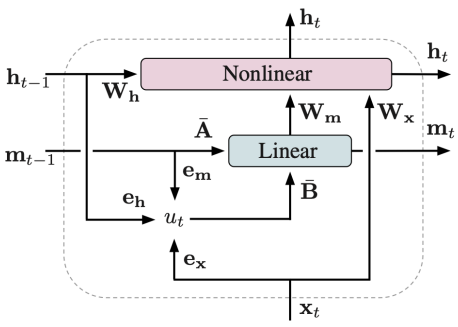


Delay Network: Windowed Sine Function



From the Delay Network to the Legendre Memory Unit

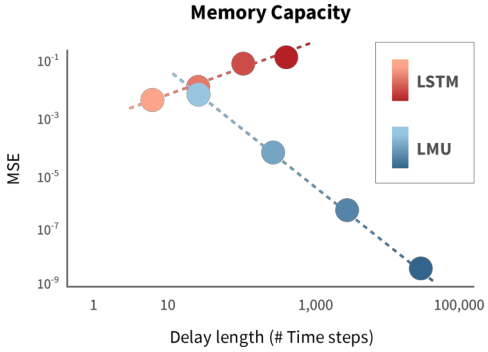
- The *Legendre Memory Unit* (LMU) places the delay network's linear system at the core of a recurrent neural architecture
- Parameter-efficient: $\mathcal{O}(N)$ rather than the expected $\mathcal{O}(N^2)$
- Trainable feedforward, deployable recurrently
- Neural connection: explains time cells; runs dynamic and spiking (or not)



State Space Models (SSMs)

- LSTMs (1995): long the dominant recurrent network, but training cannot be parallelized, and sequences beyond $\sim 10^3$ samples are out of reach
- Transformers (2017): parallel training, but compute and memory scale as $\mathcal{O}(N^2)$ in sequence length
- *SSMs*: the delay network/LMU (2018–2019, Waterloo) was the first; re-derived at Stanford as HiPPO (2020), followed by the S-series (2021) and Mamba (2023)
- SSMs now achieve state-of-the-art results on language, audio, and genomics, with SSM work at DeepMind, Meta, Nvidia, Princeton, Stanford, ...

LMU Benchmarks

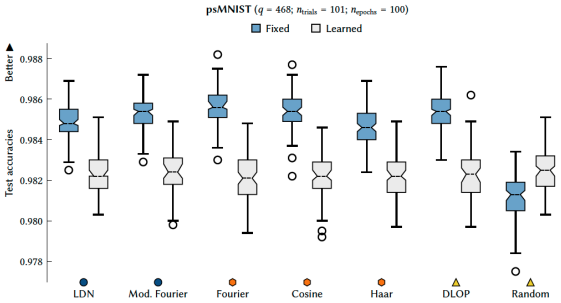


Model	Accuracy
RNN-orth	89.26
RNN-id	86.13
LSTM	89.86
LSTM-chrono	88.43
GRU	92.39
JANET	91.94
SRU	92.49
GORU	87.00
NRU	95.38
Phased LSTM	89.61
LMU	97.15
HiPPO-LegS	98.3
FF-baseline	98.20
Our Model	**98.49**

- Memory: $\sim 10^6\times$ lower error while remembering $\sim 10^4\times$ longer sequences than an LSTM
- psMNIST: (simplified) LMU set the state of the art (98.49 %) with 60 % fewer parameters
- Scaling: $\mathcal{O}(N)$ compute and $\mathcal{O}(1)$ memory, vs. $\mathcal{O}(N^2)$ for transformers

SSMs in Practice (and Other Bases)

- Applications: LLMs (e.g. Mamba), speech recognition, keyword spotting, RF classification, anomaly detection, biosignals, nonlinear system prediction
- Other temporal bases work too (Fourier, cosine, Haar, ...); analytic bases often beat learned ones



> "Next gen state space models [are] probably the next generation of transformers."
> — Jensen Huang, CEO Nvidia, 2024

Image sources

Title slide
Infrared Photograph of a Sundial Near the Einstein Tower in Potsdam, Germany
Author: DrNRNowaczyk, 2007.
From Wikimedia.